

HAMAMATSU AB, Japan

WMO: 476810

Lat: 34.750N

Lon: 137.703E

Elev: 160

StdP: 14.61

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	30.5	32.2	10.8	9.6	38.2	13.7	11.1	39.2	23.5	42.4	22.3	44.2	9.3	280	0.428

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.1	91.6	78.4	89.4	77.8	87.5	77.5	80.1	86.6	79.2	85.3	78.6	84.5	13.2	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.8	150.2	83.8	77.3	143.1	82.6	77.0	141.3	82.3	43.9	87.9	43.1	85.4	42.2	85.4	83.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
21.7	19.7	18.0	DB	27.1	96.7	1.7	2.5	25.8	98.5	24.8	99.9	23.9	101.3	22.6	103.1
			WB	23.3	82.0	1.8	0.9	22.0	82.7	20.9	83.2	19.9	83.6	18.6	84.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	61.9	42.6	44.0	49.7	58.3	66.3	72.2	79.4	81.8	76.2	67.0	56.7	47.0
	DBStd	14.17	4.17	4.97	5.30	5.34	3.73	3.88	4.01	3.11	4.40	4.63	5.24	5.23
	HDD50	617	233	178	72	5	0	0	0	0	0	0	6	123
	HDD65	2833	695	588	475	206	29	1	0	0	0	31	251	557
	CDD50	4949	3	11	62	255	505	667	911	985	786	526	208	30
	CDD65	1691	0	0	0	7	69	218	446	520	336	93	2	0
	CDH74	13146	0	0	0	22	237	955	3912	5342	2370	305	3	0
	CDH80	3940	0	0	0	1	18	170	1252	1887	581	31	0	0
(14) Wind	WSAvg	8.2	10.0	9.8	9.6	9.0	7.9	7.2	7.1	7.2	7.0	6.8	7.5	9.2
(15) Precipitation	PrecAvg	73.4	2.6	3.5	5.7	6.9	7.7	8.7	7.8	5.5	10.1	7.4	4.9	2.6
	PrecMax	100.1	6.3	8.9	9.7	13.5	12.0	14.4	16.7	16.3	25.4	27.1	9.3	5.6
	PrecMin	51.7	0.0	0.9	2.4	1.1	3.3	2.0	0.9	0.9	3.0	1.1	0.6	0.2
	PrecStd	13.1	1.6	2.0	1.7	3.0	2.6	2.7	4.2	3.3	5.2	4.7	2.4	1.6
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	58.9	64.2	69.5	77.0	82.3	89.4	95.0	96.7	91.2	83.8	73.6	65.9
		MCWB	51.1	53.9	56.1	62.1	67.1	75.8	79.1	79.2	76.3	70.9	63.1	58.0
	2%	DB	55.4	59.2	64.8	73.3	79.0	84.4	91.3	92.9	87.8	80.3	70.1	61.1
		MCWB	46.8	50.6	53.7	60.5	65.0	72.9	78.5	78.7	76.1	69.2	60.4	53.2
	5%	DB	53.2	55.7	62.5	70.0	76.8	81.0	88.2	89.8	85.8	77.2	68.1	58.8
		MCWB	44.9	47.0	53.2	59.2	64.4	71.6	78.0	78.1	75.6	67.6	59.5	50.6
	10%	DB	50.3	53.5	59.4	67.9	73.8	79.0	86.2	87.9	83.8	75.1	65.9	55.7
		MCWB	42.4	45.2	51.0	58.5	63.5	70.8	77.4	78.0	74.8	66.0	58.0	48.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.9	58.9	61.0	68.1	72.4	78.6	81.0	81.8	79.7	74.7	67.4	60.9
		MCDB	56.7	61.7	63.6	70.5	76.6	85.2	88.4	89.6	84.8	77.9	69.6	63.6
	2%	WB	49.3	53.1	58.1	65.0	69.5	76.4	79.9	80.6	78.7	72.4	63.7	55.4
		MCDB	52.5	58.1	62.2	69.3	73.7	81.2	86.7	87.2	83.6	77.0	67.7	59.6
	5%	WB	45.9	49.8	55.6	62.5	67.8	74.9	79.0	79.7	77.7	70.0	61.4	52.1
		MCDB	50.3	53.4	60.3	67.1	72.9	78.7	85.4	85.9	82.7	75.2	65.9	56.7
	10%	WB	43.6	46.7	52.9	60.4	66.0	73.1	78.2	79.0	76.5	67.9	59.3	49.8
		MCDB	49.3	51.2	58.4	65.8	71.8	76.9	84.2	84.9	81.4	73.1	64.2	54.7
(35) Mean Daily Temperature Range	5% DB	MDBR	14.3	15.5	16.3	15.9	14.6	11.7	11.3	12.1	12.3	13.4	14.4	14.3
		MCDBR	16.6	17.9	19.2	18.3	18.3	14.7	14.7	14.7	14.1	15.1	16.4	16.4
		MCWBR	11.7	12.8	13.1	11.2	9.8	7.4	6.2	6.0	6.8	8.7	10.5	11.4
	5% WB	MCDBR	14.6	16.0	15.8	14.4	13.8	11.1	12.5	12.4	11.8	12.5	13.5	14.2
		MCWBR	12.4	13.9	14.1	11.2	9.5	6.8	5.9	5.8	6.7	8.6	11.1	11.9
(40) Clear-Sky Solar Irradiance	taub		0.375	0.419	0.478	0.525	0.534	0.556	0.548	0.527	0.486	0.460	0.409	0.375
	taud		2.262	2.140	1.988	1.921	1.968	1.984	2.060	2.106	2.194	2.184	2.244	2.278
	Ebn at Noon		257	257	251	246	245	239	240	242	246	241	242	248
	Edh at Noon		34	43	54	60	58	57	53	49	43	40	34	31
(44) All-Sky Solar Radiation	RadAvg		859	1060	1348	1554	1681	1492	1658	1700	1325	1041	846	763
	RadStd		56	76	88	159	178	142	220	146	116	108	73	53

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	+0.76	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (41 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page